Fluorescent dyes

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INTRODUCTION

There are several very early reports of the unusual fluorescent brilliance of certain dyes when applied to natural textile fibres, the most pronounced effect being obtained on silk [1–4]. The greenish-yellow fluorescein and its orange to red halogenated derivatives (eosins) were among the dyes described in early literature, but their practical use was limited by inadequate fastness properties. The brilliant bluish-red rhodamines were found to exhibit rather better fastness properties on textiles and they remain to this day one of the most important classes of fluorescent dyes for a wide range of applications.

The search for new fluorescent dyes for textile applications gained impetus around the middle of this century from the introduction of synthetic fibres, notably polyester, nylon and acrylics. Around the same time the development of daylight fluorescent pigments, which impart a vivid brilliance to a range of paint and printing ink applications, provided a requirement for dyes with improved durability. Research aimed at producing new fluorescent dyes for use in the coloration of textiles, coatings and plastics has continued in recent years with the objectives of extending the available product shade range, enhancing fluorescence intensity and improving other useful application properties such as light fastness and thermal stability. In addition, several new applications have emerged which specifically exploit the light emission properties of fluorescent dyes. These functional applications include non-destructive flaw detection, tunable dye lasers, solar collectors, liquid crystal displays and electroluminescent displays, and the search for new dyes with appropriate properties for such uses appears to be continuing unabated. Techniques involving the use of specific fluorescent dyes are of immense importance in biological and medical research, providing a useful tool in the search for new biologically active compounds and the potential for the development of new diagnostic methods.

It is difficult to arrive at a precise definition of what constitutes a fluorescent dye. Many coloured substances exhibit fluorescence to a certain degree. Azo dyes, by far the most important chemical class of colorant, are generally non-fluorescent. However, it is probable that certain dyes from other classes of traditional colouring materials, including anthraquinones, phthalocyanines and the more recently introduced benzodifuranones, owe their brightness of colour to a certain extent to a small but visibly detectable fluorescence.

For the purposes of this review, fluorescent dyes have been defined as materials that both absorb and emit strongly in the visible region of the spectrum and that owe their potential for application principally to their intense fluorescence. This definition is somewhat more restrictive than that used by many other authors since it excludes fluorescent brightening agents, i.e. compounds which emit visible light but absorb in the UV region. There are certainly important similarities between fluorescent brightening agents and fluorescent dyes as defined in this review, especially in terms of the fluorescence mechanism, the structural units of many of the chromophores used, and the applications. However, a practical reason for the exclusion of fluorescent brightening agents is that those materials have been the subject of extensive previous reports including two reviews in this journal [5,6]. Additionally the definition also excludes compounds that absorb visible light but emit in the near-infrared region. Such materials are of potential commercial importance, particularly for application in tunable near-infrared dye lasers.

In this article the structures and properties of the most important classes of fluorescent dyes are discussed extensively. The subject is too vast a subject, especially for specialist applications, to attempt to be comprehensive. In addition an overview of their applications is presented. To keep the article to a reasonable length, synthetic routes to the dyes are not covered, but the reader will be able to obtain details of these from the references provided.

The field covered is not particularly well served by previous review articles. There is, however, an excellent book that provides a comprehensive account of organic luminescent materials, with particular emphasis on research carried out in the USSR [7]. Other articles deal with the application of fluorescent dyes as components of daylight fluorescent pigments [8–10].

FLUORESCENCE

Detailed theoretical aspects of fluorescence have been covered extensively in the literature [10–17]. The energy transitions that occur in a molecule exhibiting fluorescence are illustrated in Figure 1. When the molecule absorbs light, it is excited from the lowest vibrational level in its ground state (S_0) to a range of vibrational levels in the singlet first excited state (S_1^*). In the case of organic fluorescent dyes, this is generally a π–π^* electronic

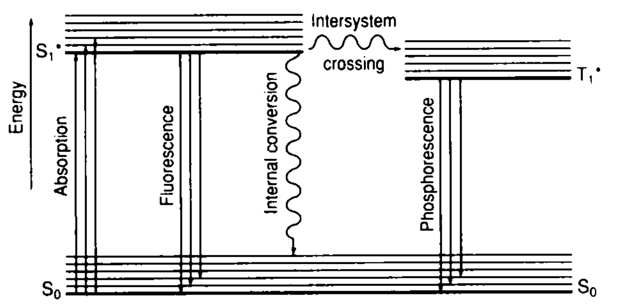

Figure 1 Some energy transitions in a fluorescent dye

transition. During the time the molecule spends in the excited state, energy is dissipated from the higher vibrational levels, and the lowest vibrational level is attained. Fluorescence occurs if the molecule then emits light as it reverts from this level to various vibrational levels in the ground state. Non-radiative processes, the most important of which is generally collisional deactivation, also give rise to dissipation of energy from the excited state. As a result there will be a reduction in the intensity of fluorescence and in many cases it will be absent altogether.

Another process which may occur is intersystem crossing to a triplet state. Emission of light from the triplet state is termed phosphorescence, a phenomenon which is outside the scope of this review. As a consequence of the loss of vibrational energy in the excited state, fluorescent emission occurs at longer wavelengths than absorption. The difference between the emission and absorption wavelength maxima is referred to as the Stokes shift. The intensity of light emission from a fluorescent dye is characterised by its quantum yield, which is the ratio between the emitted and absorbed quanta. In some fluorescent materials this may approach unity. In the case of a fluorescent dye as defined in this review, the overall visual effect from the dye, whether in solution or incorporated into a textile fabric, plastic material or surface coating, originates from the colour attributable to the selective absorption of light supplemented by the colour of the light emitted.

REQUIREMENTS OF FLUORESCENT DYES

Fluorescent dyes are required to absorb in the visible region and to exhibit intense emission. Additional characteristics, such as appropriate fastness properties, functional properties, non-toxicity, etc., will be dictated by particular applications. The relationship between the colour of organic dyes, when this is due only to light absorption, and their molecular constitution is now well understood in qualitative terms, and an excellent quantitative treatment has emerged as a result of the application of quantum mechanical methods, notably the PPP MO approach [18–21]. At present the relationship between the molecular constitution and emission properties of dyes is less completely understood. The magnitude of the Stokes shift is an important parameter for certain applications of fluorescent dyes. For example, in dye lasers, solar collectors and certain biological applications a large Stokes shift is desirable to minimise the overlap between the absorption and emission spectra. The magnitude of the Stokes shift in fluorescent dyes is generally around 50–70 nm. In certain dyes the shift may increase to 150–250 nm. This may be observed, for example, when a molecule in the first excited state adopts a more coplanar configuration than in the ground state or when intramolecular hydrogen bonding becomes stronger after excitation.

Some progress has been made in establishing a semi-empirical approach to the calculation of Stokes shifts in fluorescent dyes, in particular by the use of a modification to the PPP MO method [22–26]. In this approach it is assumed that ground-state geometries (e.g. from X-ray crystallographic data) may be used in the calculation of absorption maxima, while the position of fluorescence maxima is determined by excited-state geometries. A conventional PPP MO calculation is carried out on the dye in question, using standard bond lengths and resonance integrals, thus giving a predicted absorption λ_{max} value. In addition, the π-bond orders for the first excited-state are calculated. Subsequently, from the excited-state bond orders, excited state bond lengths and resonance integrals are calculated using established mathematical relationships. A second PPP calculation is then carried out using these new parameters. In an iterative procedure these calculations are repeated until convergence of bond lengths or transition energies is achieved. The new λ_{max} value that results is equated to the fluorescence maximum and hence the Stokes shift is obtained as the difference between this value and the original result. This method lacks rigorous theoretical justification but good results have been reported for a range of fluorescent dyes, including coumarins and naphthalimides.

The relationship between the molecular structure of organic dyes and the intensity of their fluorescence is by no means well established. Intense fluorescence is associated with the presence of certain structural units, and these form the basis of the classification of fluorescent dyes used in this review. Fluorescent dyes are frequently extended conjugated systems, often containing a number of fused rings. An important structural feature that almost invariably leads to enhanced fluorescence quantum yields is increased structural rigidity; this minimises loss of energy from excited states by intramolecular thermal motion. The reduction of fluorescence intensity in molecules lacking in structural rigidity has been termed, by an appropriate mechanical analogy, the 'loose-bolt effect'.

An important challenge remaining to be tackled successfully, yet which would be of immense value in the design of new fluorescent dyes, is the development of a quantitative treatment of quantum yields. This will probably require a theoretical approach to the relative rates of fluorescence emission and competing non-

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radiative deactivation processes from the excited state. A notable difficulty inherent in establishing such structure–property relationships is the significant influence on fluorescence emission spectra of environmental effects, such as the nature of the solvent, concentration and temperature.

FLUORESCENT CARBONYL DYES

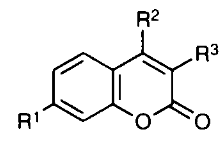
An extensive range of fluorescent dyes that contain the carbonyl group as an essential feature are classified together in this section.

Coumarins and related compounds

Coumarins, in addition to their importance as fluorescent brightening agents, represent one of the most extensively investigated and commercially significant groups of fluorescent dyes. Although the most important commercial coumarin dyes are yellow with a green fluorescence, fluorescent coumarins have been reported which absorb and emit in most parts of the visible spectrum. The parent molecule (**1a**) is a colourless material that does not fluoresce at normal temperatures. The important fluorescent coumarin derivatives almost invariably contain an electron-releasing substituent in the 7-position, which is most commonly a dialkylamino group but may be hydroxy or methoxy. Additionally it has been noted that electron-withdrawing substituents in the 3- and 4-positions, such as the cyano group, lead to further shifts of the absorption and emission bands to longer wavelengths. For example, compound **1b** is colourless while **1c** is yellow [27]. The colours exhibited by coumarins may be explained on the basis of typical donor–acceptor behaviour.

The earliest and still the most widely used coumarin dyes contain a benzimidazolyl (**2a**), benzoxazolyl (**2b**) or benzothiazolyl (**2c**) group as the acceptor in the 3-position [28–34]. These dyes were found to enable synthetic fibres such as polyester to be dyed in greenish-yellow shades possessing a previously unattained brilliance. In the series the thiazole derivative **2c** is the most bathochromic. CI Disperse Yellow 232 has been disclosed as the chlorobenzoxazolyl derivative **2d** [35], and is reported to have a brilliant greenish-yellow hue on polyester with good fastness to light, sublimation and washing. The use of these coumarin dyes has been extended subsequently to a much wider range of applications including daylight fluorescent pigments, light collector systems and dye lasers.

Following the discovery of this group of dyes, there were some reports in which new coumarin derivatives,



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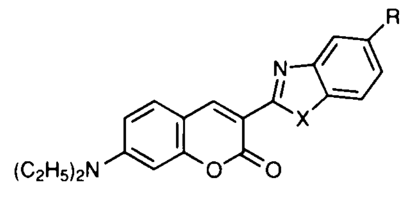
	R^1	R^2	R^3
1a	H	H	H
1b	OCH_3	H	H
1c	OCH_3	CN	CN

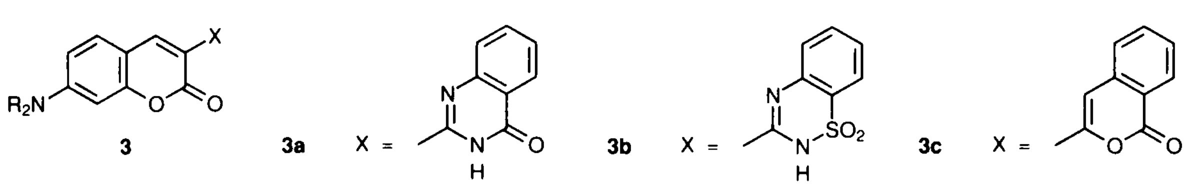


2

	X	R
2a	NH	H
2b	O	H
2c	S	H
2d	O	Cl

based mainly on variations of the heterocyclic substituent in the 3-position, were suggested. For example, dyes **3a** [36,37], **3b** [38] and **3c** [39] are reported to be suitable for dyeing synthetic fibres and some plastics in greenish-yellow shades with good fastness to light and migration. The other dyes of this type reported include pyrazolyl [40], furyl, oxazolyl and oxadiazolyl [41], imidazolyl and pyrimidyl [42], quinolinyl [43] and thiazolyl and thiadiazolyl [44] derivatives. Several structurally related cationic coumarin dyes are also known. These include the quaternised pyridine derivative **4** [45], reported to impart bright orange hues to acrylic fibres, and compound **5** [46], which provides fast greenish-yellow shades on acrylics and acid-modified polyesters.

Benzazolyl-substituted coumarins can also be converted into water-soluble cationic dyes by alkylation of the azole ring nitrogen atoms [47,48]. The most important commercial product of this type is CI Basic Yellow 40 (**6**), a greenish-yellow which finds applications in acrylic fibres and in daylight fluorescent pigments [9]. Other fluorescent cationic coumarin dyes have been reported, invariably involving heterocyclic groups in the 3-position containing quaternised nitrogen [48,49]. The laser



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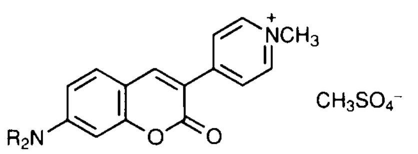

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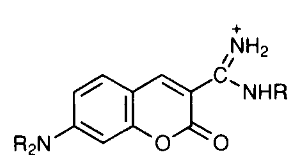

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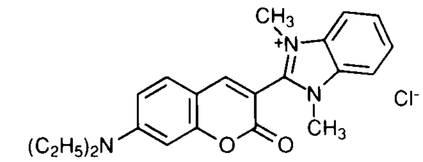
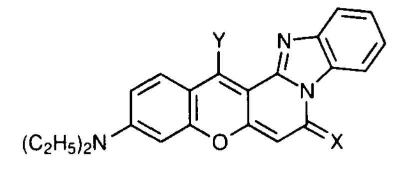


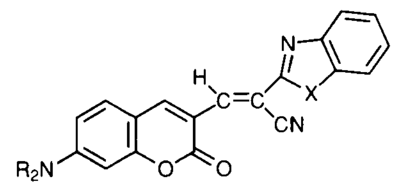
6

characteristics of a range of 3-hetarylcoumarin dyes have been studied [50–52]. A series of compounds **7** (X = NH, O or S), described as having 'butterfly' structures as a result of the julolidine ring system, provide quantum yields approaching unity, an example of the enhanced fluorescence efficiency arising from a more rigid molecular structure [51]. The fluorescent colours exhibited by the series of dyes **2–7** are restricted essentially to yellows and oranges. However, it has been shown that introduction of a cyano group into the 4-position gives rise to a significant bathochromic shift producing useful fluorescent red dyes [27,53–57]. For example, the absorption maximum of the benzthiazolyl dye **8** in acetone is caused by the cyano group to shift to longer wavelengths by 67 nm, while the fluorescence maximum on dyed polyester is displaced bathofluorically by 115 nm.



9

9a X = O; Y = H
9b X = NR; Y = H
9c X = NH; Y = CN



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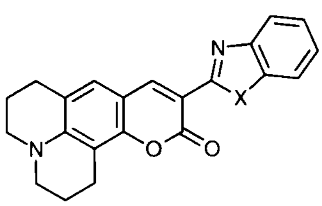

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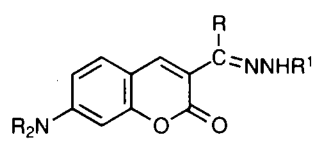

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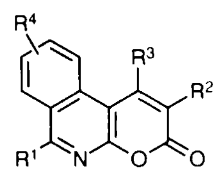

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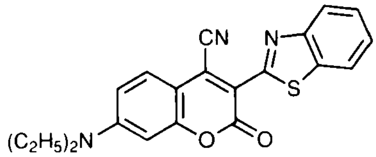

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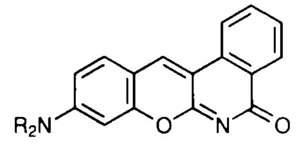
Other examples of coumarin dyes absorbing and emitting at significantly longer wavelengths are **9a** and **9b**, formed as a result of further ring closure. These are fluorescent red dyes suitable for application to polyester [58–60], while the cyano derivative **9c** is reported as a greenish-blue dye exhibiting red fluorescence (absorption maximum at 630 nm, emission maximum at 668 nm) [27].

Compounds of type **10** (X = NH, O or S) have been reported as fluorescent red dyes suitable for dyeing and transfer printing on polyester [61], the bathochromic shift resulting from extended conjugation. Some coumarin hydrazones (**11**) have been described as intense yellow compounds in solution exhibiting an intense bluish-green fluorescence and have been investigated as photostable laser dyes [62]. A range of azacoumarins (**12**) are reported as brilliant greenish-yellow to blue dyes, depending on the substituent pattern, and are suitable for dyeing polyester [63].

Other fused-ring heterocyclic coumarin analogues reported include the fluorescent orange **13** [64,65], suitable for dyeing polyester and certain plastics, while the structurally more complex derivatives **14** give fluorescent greenish-yellow shades on polyester with moderate light fastness and good sublimation fastness [66]. The synthesis of an extensive range of benzoxazinones, which may be considered as 4-azacoumarins, has been reported [67–69]. Many of the compounds exhibit strong fluorescence both in solution and the solid state. However, they appear to lack in general the stability necessary for commercial exploitation.

The formylstyryl derivative **15**, suggested as the most promising member of the series, is a red fluorescent material exhibiting a large Stokes shift (approx. 110 nm) in polar solvents and has been investigated for its potential



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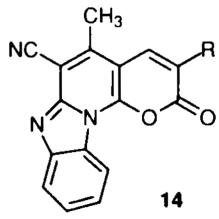
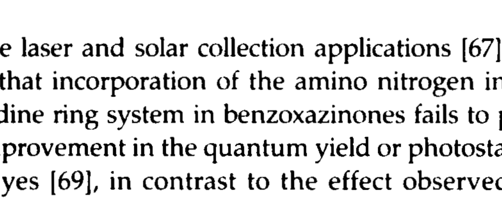


14

in dye laser and solar collection applications [67]. It is of note that incorporation of the amino nitrogen in a rigid julolidine ring system in benzoxazinones fails to produce an improvement in the quantum yield or photostability of the dyes [69], in contrast to the effect observed in the coumarins [51]. Other coumarin derivatives that have been reported as yellow to red fluorescent dyes suitable for application to polyester include fused imide and anhydride derivatives **16** [70], azomethine derivatives **17a** [71,72] and the methine derivatives **17b** [73], in which the group Y may be a cyano, ester or amide group.



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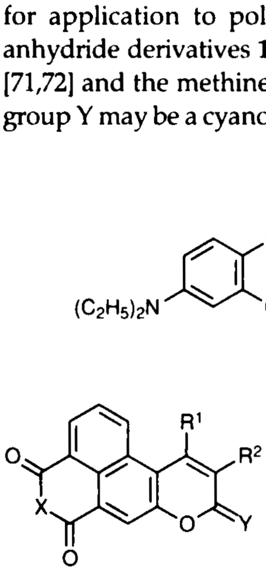
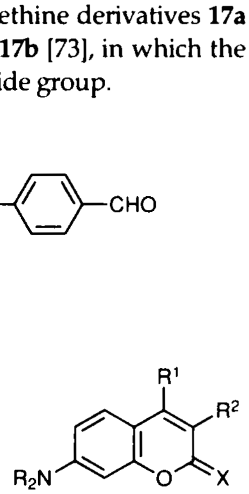


16 X = O, NR
Y = O, NR

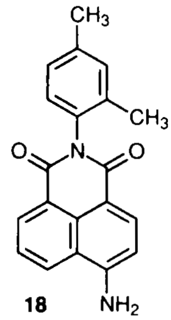


17

17a X = NH, NR³
17b X = CY(CN)

Naphthalic acid derivatives

Fluorescent materials based on naphthalic acid (naphthalene-1,8-dicarboxylic acid) and its anhydride are known. However, it is the imides derived from this acid that provide the most useful fluorescent brightening agents and dyes. Strong fluorescence is observed in such derivatives when there is an electron-donating group in the 4-position, in which case the absorption and emission properties are associated with intramolecular charge transfer involving the donor group and the electron-withdrawing *peri*-carbonyl groups. Of greatest commercial significance in this series are the 4-aminonaphthalimides. In particular CI Disperse Yellow 11 (**18**), which contains an unsubstituted amino group and an aryl substituent on the imide nitrogen, is a long-established fluorescent yellow dye that can be used in daylight fluorescent pigments and in the coloration of synthetic fibres and plastics, although it possesses only moderate light fastness [74]. The related compound **19**, containing n-butyl groups on both the imide and amine nitrogens, is appreciably redder and stronger than **18** but exhibits poorer light stability.



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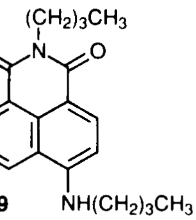

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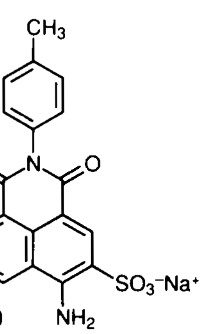
The principal variation on the structure of aminonaphthalimides that has been investigated is the nature of the substituent on the imide nitrogen. A wide range of such substituents have been examined including alkyl, aryl, aralkyl, alkoxy and amino groups [75–80]. In general the nature of the substituent is observed to have little effect on the absorption maximum but can have a somewhat more pronounced influence on the emission maximum. A reasonable correlation between the experimental Stokes shift for a series of aminonaphthalimides and that calculated by a modification of the PPP MO method has been reported [26]. Several water-soluble naphthalimide dyes are known including the anionic dyes CI Acid Yellow 7 (**20**) and CI Mordant Yellow 33 (**21**) [81], which have been used as dyes for silk and may also be used in flaw detection. There are also patent reports of cationic aminonaphthalimide dyes, such as compounds **22** [82] and **23** [83], described as suitable for dyeing acrylic fibres in fluorescent greenish-yellow shades. Fluorescent yellow reactive dyes for cellulosic fibres, such as **24**, have also been described [84].

Compounds **25** [85] and **26** [86] are examples of useful dyes in which the amino nitrogen has been condensed into a heterocyclic ring system. The aminonaphthalimide dyes described thus far in this section are invariably



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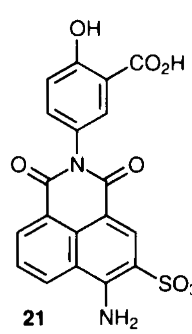

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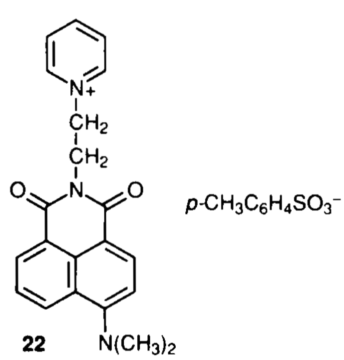

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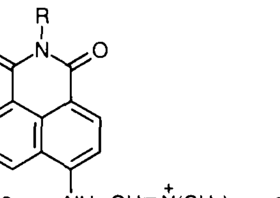

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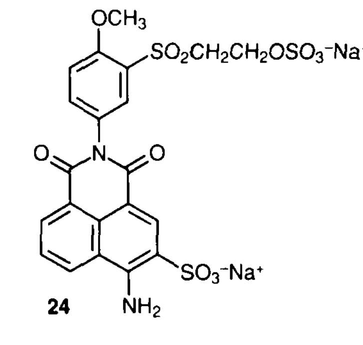

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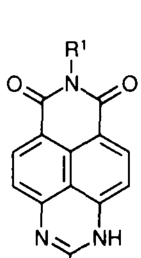

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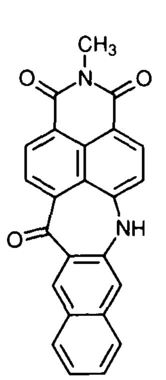

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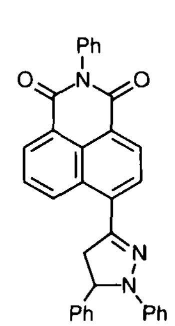

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restricted to the yellow shade area, so it is of interest that some pyrazoline derivatives, such as **27**, allow extension of the shade range into the orange-red region [87].

A series of related naphthalic acid derivatives that offer some potential as fluorescent dyes are the 1,8-naphthoylene-1′,2′-benzimidazoles. The parent compound **28** exhibits intense yellow-green fluorescence both in hydrocarbon solvents and in the solid state [88,89]. Its use has been reported in flaw detection, in the fluorescent dyeing of polyester and other polymers [90], and in biological applications [91]. A wide range of substituted compounds are also known [26,75,92–97]. Electron-releasing (e.g. amino, alkoxy) substituents in the 4- and 5-positions of the naphthalene ring system give rise to significant bathochromic and bathofluoric shifts with respect to the parent compound, allowing extension of the shade range from greenish-yellow into the orange–scarlet region. A reasonable correlation between the experimental Stokes shift for a series of these dyes and that calculated by a modification of the PPP MO method has been reported [26].

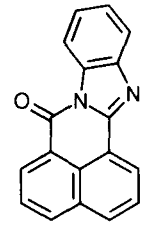
Quaternised derivatives, such as **29**, derived from methylation of the parent compound, are described as useful cationic dyes for acrylic fibres, compound **29** giving a bright greenish-yellow light-fast colour [98]. A range of related compounds that contain additional carbocyclic or heterocyclic ring systems are known, including compounds **30–34**, providing greenish-yellow to yellowish-red fluorescent dyes for polyester and plastics [99–104].

Dyes such as **35**, which may also be considered as naphthalic acid derivatives, are reported to dye polyester in fluorescent greenish-yellow shades that are fast to light and sublimation [105].

Perylenes

Perylene (**36**) is an aromatic hydrocarbon with interesting luminescence properties, particularly for laser applications [106]. When excited by a laser, it gives a lasing maximum at 470–480 nm, the lasing band being 1 nm wide. Since the most intensely fluorescing perylene dyes contain carbonyl groups, they are discussed in this section.

The diisobutyl ester of perylene-3,9-dicarboxylic acid (**37a**, CI Solvent Green 5) exhibits strong yellow-green luminescence and is a long-established dye for use particularly in flaw detection. A feature of compounds of this type, especially some derivatives containing more rigid bicycloalkyl ester functionality such as compound



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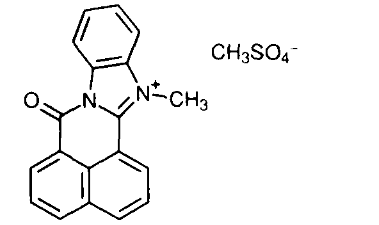

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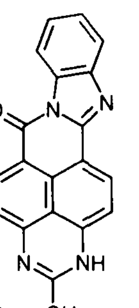

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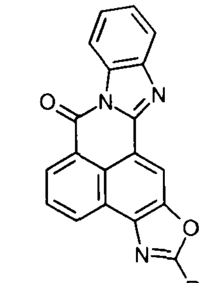

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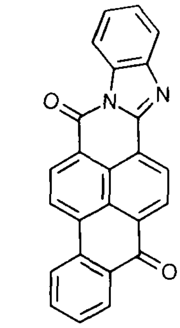

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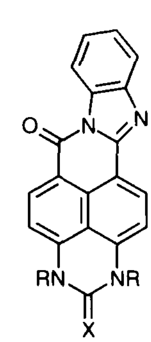
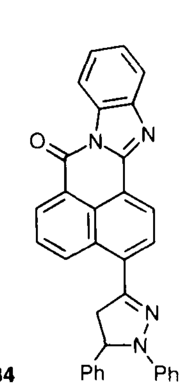


33 X = O, S



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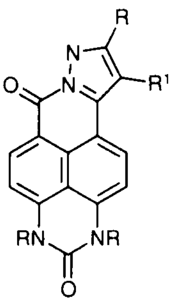

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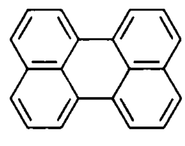
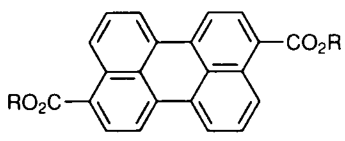


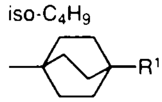
36

37b, is their application as pleochroic fluorescent dyes in liquid crystal display systems [107]. The linear, rod-like nature of the molecules is an important structural feature for this application. When dissolved in a nematic host material, the long molecular axis of the dye aligns with the director of the liquid crystal, and control of dye absorption and emission may be achieved by altering the alignment of the liquid crystal when an external electric field is applied. A range of esters of tri- and tetra-carboxylic acids of perylene have also been suggested as fast fluorescent yellow to red dyes for the coloration of polyester fibres and styrene polymers [108], as have a series of analogous thioesters [109], and the use of cyano derivatives in solar energy collection has been reported [110].



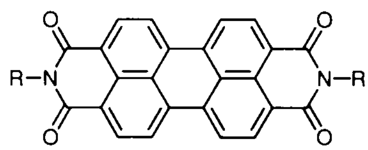
37

37a R = iso-C_4H_9

37b R = 

Recent years have seen extensive research activity centred on a range of *NN*-disubstituted diimides of perylene-3,4,9,10-tetracarboxylic acid, which provide intense orange to red fluorescent colours in polymers. These perylene dyes are especially significant as they provide bright colours with very high fluorescence efficiency while at the same time being extremely stable towards light and heat, and resistant to chemical attack. For these reasons they are attractive materials for the coloration of thermoplastics [111], and they are frequently the materials of choice for 'high tech' applications such as liquid crystal displays [112–115] and solar energy collection [116–120] where durability is of prime importance.

Many perylenediimides are extremely insoluble materials and are of great importance as high-performance pigments. Compound **38a** provides an example of an important group of fluorescent dyes in which the imide nitrogens are substituted by aryl groups with bulky alkyl functionality in the *ortho* positions [121–123]. The tertiary butyl groups in **38a** ensure solubility in organic solvents, and additionally the bulky nature of the groups sterically prevents rotation about the C–N bond. The compound has in fact been obtained in two atropisomeric forms that have been shown to be quite stable at elevated temperatures in solvents without any interconversion [124]. Compounds of this type can give quantum yields approaching unity, and it has been generally assumed that this is associated with rigidity of the molecules arising from the conformational restriction. However, it has been observed more recently that dyes with secondary alkyl substituents on the imide nitrogens, such as **38b**, also exhibit fluorescent quantum yields of about unity in solution even though the rotational barrier in these cases is rather lower. This suggests that conformational mobility may not be the only factor influencing the fluorescence deactivation processes [125,126].



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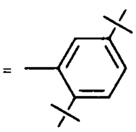
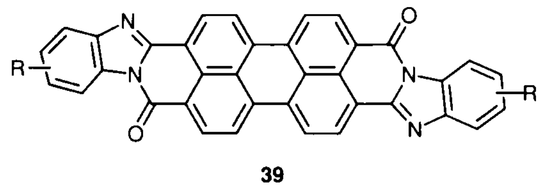
38a R = 

38b R = —$CH(C_6H_{13})_2$

Water-soluble perylenediimides have been prepared by incorporating sulphonic acid groups onto aryl substituents on the imide nitrogens [127]. Asymmetrically substituted perylenediimides have been synthesised and their use suggested as fluorescent tracers in biochemistry [128,129]. Related to the diimide dyes are bisbenzimidazole perylene derivatives, such as **39**, which give light fast fluorescent reddish-violet colours when incorporated into thermoplastics [130].



39

Benzanthrone derivatives

The longest established fluorescent benzanthrone dye is the 3-methoxy derivative **40a** (CI Disperse Yellow 13), which exhibits yellow-green fluorescence in the solid state and in solution [131,132]. This molecule represents a further example of a donor–acceptor chromogen, the absorption and emission properties being associated with a conjugated interaction between the electron-releasing methoxy group and the carbonyl group. The 2-methoxy

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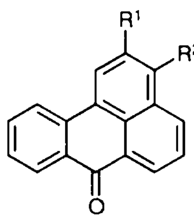
isomer **40b**, which has no pathway of conjugation between the methoxy and carbonyl groups, is not fluorescent. Compound **40a** provides a relatively inexpensive means of producing fluorescent greenish-yellow colours, but it suffers from the disadvantage of a rather low quantum yield. However, the fluorescence intensity of the dye can increase when incorporated into certain resin systems used in daylight fluorescent pigments, an effect attributed to intermolecular hydrogen bonding between dye and resin molecules. In addition, when used in conjunction with certain fluorescent brightening agents, an enhancement of fluorescence intensity can occur as a result of intermolecular energy transfer [133]. 3-Aminobenzanthrone (**40c**) is a fluorescent orange compound with limited commercial importance [134].



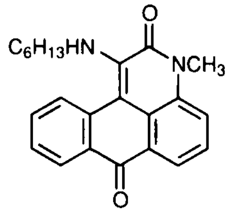
40

	R^1	R^2
40a	H	OCH_3
40b	OCH_3	H
40c	H	NH_2

Analogues of the benzanthrone system have been investigated in attempts to achieve a higher quantum yield than is obtainable from the simply substituted benzanthrone derivatives. For example, 1-hexylamino-*N*-methylanthrapyridone (**41**) is an efficient yellow-green luminophor with a quantum yield much higher than that of compound **40a** [135–137]. The pyrimidanthrone derivatives, such as **42a** and **42b**, are yellow to reddish-orange colorants used in the coloration of plastics [7–10]. Azabenzanthrone dyes, such as **43**, are suggested as providing fluorescent yellows in polymethylmethacrylate with good light fastness and heat stability [138], while the polycyclic azapyrene derivatives **44** are fluorescent violet dyes [139].

Anthraquinones

Some anthraquinone dyes exhibit a weak fluorescence responsible at least in part for their brightness of colour when applied to textile fibres. These dyes contain hydroxy or amino groups in the α-positions capable of intramolecular hydrogen bonding with the carbonyl groups. A few anthraquinone derivatives, generally with extended rigid structures, exhibit more pronounced emission and may be considered as fluorescent dyes within the scope of this review. Dialkylamino-substituted anthraquinonediazole derivatives **45** (X = O, S or Se) have red or violet colours and luminesce at long wavelengths with quantum yields up to 0.66 [140,141]. These dyes find



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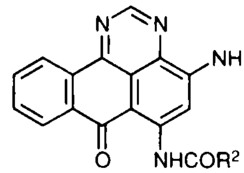
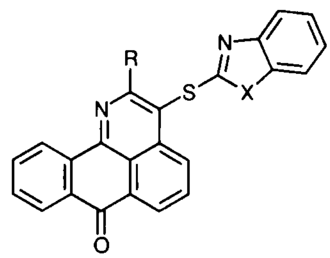
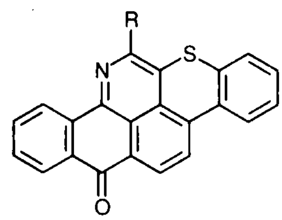


42

	R^1	R^2
42a	phenyl	2,5-dichlorophenyl
42b	$C_{18}H_{37}$	C_4H_9



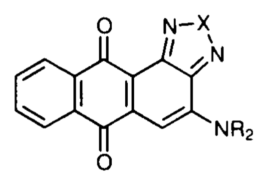
43 X = NH, O, S



44

some use in the manufacture of violet daylight fluorescent pigments. A group of anthraquinonoxazoles have been proposed as fluorescent yellow to red dyes for synthetic fibres and plastics, including the scarlet dye **46** [142].

Dichloroisoviolanthrone (**47**), which may be used as a vat dye for cellulosic fibres, has been described as imparting reddish-violet fluorescent colours to polystyrene and polymethylmethacrylate [143], while some related compounds containing long-chain alkyl substituents have been suggested for application in solar light collection [144].



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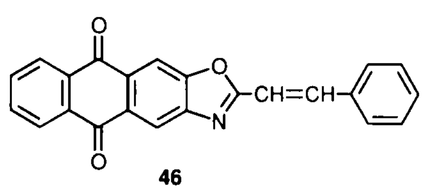

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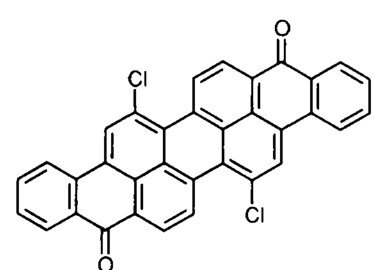

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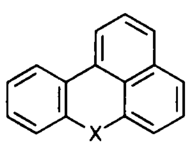
Benzoxanthones and benzothioxanthones

Following a period of extensive research into a range of dyes based on the benzoxanthene (**48**, X = O) and the benzothioxanthene (**48**, X = S) systems, Hoechst introduced some brilliant fluorescent and light fast benzothioxanthones as disperse dyes for polyester during the 1970s [145–160]. A range of fluorescent colours from yellow through orange into the bluish-red region has been obtained by variation of the substitution pattern or by ring annellation. CI Disperse Red 303, a fluorescent yellowish-red, the only commercial product of the series whose structure has been disclosed to date, has been shown to exist as a mixture of isomers **49** and **50** [161]. It is interesting that these two isomers show significantly different absorption and emission characteristics. Compound **49** is orange with a reported λ_{max} in methanol of 499 nm, while isomer **50** is red with a λ_{max} of 511 nm. The orange component exhibits three times the fluorescence intensity of the red component.

Compounds containing naphthalimide functionality, exemplified by the greenish-yellow dye **51** and the bluish-red dye **52**, are also widely reported in the patent literature. In addition to their textile applications, derivatives of this type have been suggested for use in liquid crystal display systems [162]. Some sulphonated derivatives are useful as water-soluble fluorescent yellow dyes for inks [163].



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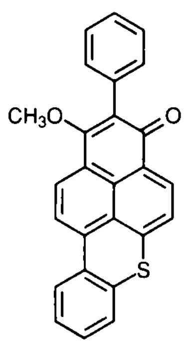

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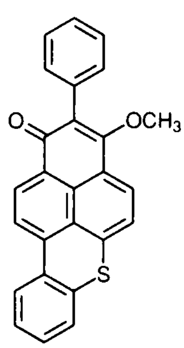

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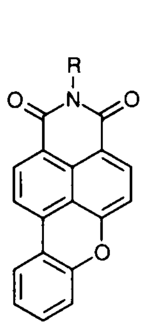

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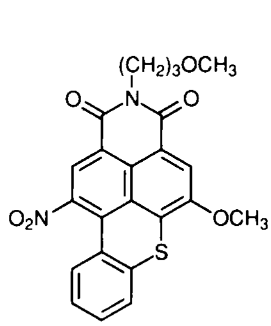

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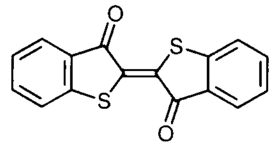

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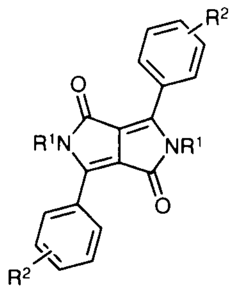
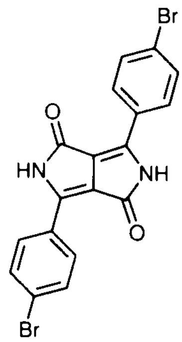


54 R^1 = H, CH_3



55

Other fluorescent carbonyl dyes

Thioindigo (**53**) and some halogenated and alkylated derivatives, although more commonly employed as vat dyes or pigments, are also used, often in combination with other luminophors, to impart fluorescence. They impart red to violet colours in certain thermoplastics, notably polystyrene, polymethylmethacrylate, polyesters and PVC, where they offer some resistance to heat and light [164–167].

Arguably the most important development in organic pigments in recent years has been the discovery and commercial introduction of the bright red diketopyrrolopyrrole (DPP) pigments. In a recent publication, a range of 3,6-diaryl DPP derivatives (**54**) containing alkyl substituents, e.g. a tertiary butyl group, on the aryl rings are reported as fluorescent orange to red dyes with Stokes shifts of up to 75 nm and quantum yields up to 0.95. In addition they exhibit good solubility in organic solvents and high photostability [168]. These results demonstrate considerable potential for a range of applications. A patent describes the bromo derivative **55** as a fluorescent red pigment suitable for use in the coloration of polymers [169], while the use of DPP-based solid-state fluorescent dyes in optical memory devices has been reported [170].

Dyes derived from maleic anhydride containing further heterocyclic substituents include compound **56**, described as imparting a reddish-yellow colour to polystyrene and an intensely fluorescing orange to polyester fibres [171]. 2,5-Diarylthiophen-3,4-dicarboximides (**57**) have been patented as yellow fluorescent dyes suitable for polyester fibres [172].

Bisoxazolones (**58**) have been proposed as greenish-yellow fluorescent dyes for polyester fibres [173–176].

RHODAMINES AND RELATED COMPOUNDS

Triarylmethane dyes do not as a rule exhibit strong fluorescence. However, certain related dyes that contain a

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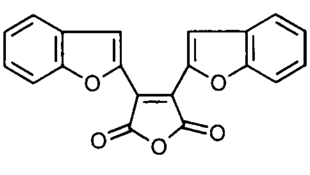

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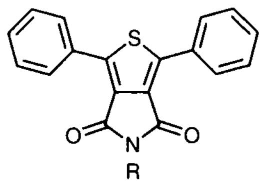

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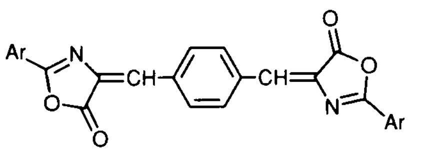

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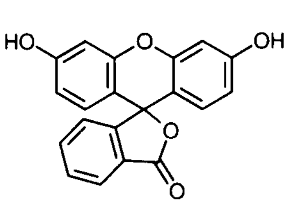
pyran ring are strongly fluorescent, presumably as a result of the enhanced molecular rigidity due to the oxygen bridge. This group of compounds includes the 'classic' luminophors fluorescein and the rhodamines. Fluorescein (**59**) and uranine (**60**), which is the product of alkaline ring opening of fluorescein, are yellow dyes with an intense green fluorescence [177]. They are no longer of any real significance as traditional colouring materials but they are used extensively in hydrological tracing and in analytical and biological applications.

Of major importance as fluorescent dyes are the rhodamines, which are structurally related to the ring-opened form of fluorescein but have amino groups in place of the hydroxy groups, and are intensely fluorescing red to violet materials. Rhodamine B (**61a**, CI Basic Violet 10) and rhodamine 6G (**61b**, CI Basic Red 1) are particularly important, and both have a wide variety of traditional and 'high tech' applications. Rhodamine 6G is an important red dye for use in daylight fluorescent pigments and was one of the first to be used in dye lasers.



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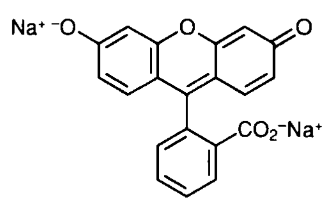

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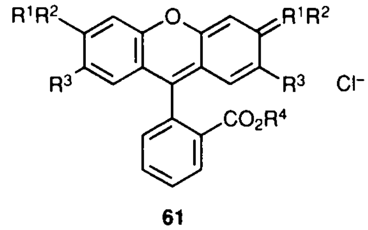
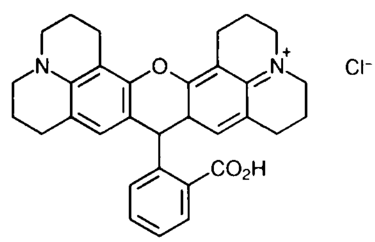


61

	R^1	R^2	R^3	R^4
61a	C_2H_5	C_2H_5	H	H
61b	H	C_2H_5	CH_3	C_2H_5

It is probably still the most commonly used compound in dye lasers, its behaviour characterised by a high transformation coefficient, a low lasing threshold and invariability of optical behaviour in the case of repetitive pumping by powerful flashlamps [178,179].

The substituents on the amino groups have been shown to exert an important influence on the absorption and emission properties of rhodamines. In general alkyl substituents cause bathochromic and bathofluoric shifts and raise the quantum yield. Aryl substituents shift the absorption and emission to longer wavelengths but reduce fluorescence intensity. Some asymmetric rhodamines have been investigated but in general these show lower quantum yields than the symmetrical derivatives [180]. In rhodamine 6G the pendant ester group prevents free rotation of the lower phenyl ring, an effect which probably contributes towards its fluorescence efficiency. The effect of forming a more rigid structure in fluorescent dyes of the rhodamine series has been demonstrated in the case of the remarkable dye **62**. The incorporation of the two nitrogen atoms in rigid rings leads to a quantum yield close to unity and independent of temperature [181,182].



62

A feature of the rhodamine series of dyes is their susceptibility towards concentration quenching in solution. This limits the concentration at which they may be used effectively in solid solutions in resins to form daylight fluorescent pigments. The effect is attributed to the formation of non-fluorescent dye aggregates, the extent of which increases as the concentration is raised [183]. In spite of the fact that rhodamines have been known and used for many years, there remains several incompletely explained features of their fluorescence characteristics, particularly aggregation and vapour-phase phenomena, and no doubt these will continue to attract the attention of researchers [184,185].

Acridine dyes may be considered as heterocyclic nitrogen analogues of the rhodamines. Examples of these are acridine yellow (**63a**), acridine orange (**63b**), acridine orange R (**63c**) and proflavine (**63d**). These dyes are strongly fluorescent but, like fluorescein and its derivatives, they are of inadequate stability for most traditional coloration applications. They are, however, of some significance as indicators in acid/base titrations and also as fluorescent probes because of their ability to intercalate with biological macromolecules [186]. Some quaternised acridinium salts and related heterocyclic nitrogen derivatives have been investigated as biological

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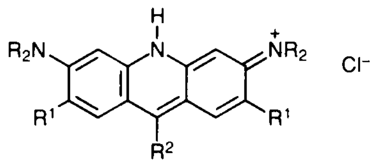
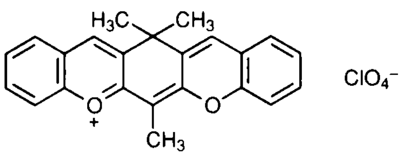
probes whose fluorescence is relatively independent of pH [187].

Compound **64** is described, together with a wide range of related cationic heterocyclic derivatives, as a fluorescent red dye suitable for dyeing anionically modified fibres [188].



63

	R	R¹	R²
63a	H	CH_3	H
63b	CH_3	H	H
63c	CH_3	H	Ph
63d	H	H	H



64

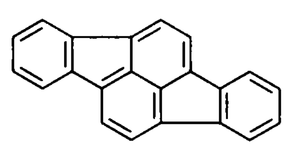
HYDROCARBON AND METHINE FLUORESCENT DYES

Many polycyclic aromatic hydrocarbons display strong fluorescence in the visible region, but their commercial exploitation is restricted by difficulties in synthesis and their potentially carcinogenic nature in some cases. The use of polycyclic hydrocarbons in the coloration of polyester, nylon and polyethylene has been proposed in patents. For example, rubicene (**65**) is reported as producing bright orange colours in these polymers [189], while hydrocarbons of the type **66** are reported to exhibit intense yellow fluorescence in injection-moulded polyester [190]. The pyrene derivative **67** (CI Solvent Green 7) is an established textile dye, displaying yellow-green fluorescence [191].

Stilbene (**68a**) and its derivatives are well known fluorescent compounds and they find extensive application as fluorescent brightening agents. Coloured fluorescent stilbene derivatives are less well known. However, 4-dimethylamino-4′-nitrostilbene (**68b**), which has donor–acceptor character, is reported to give an absorption maximum in benzene at 430 nm, a fluorescence maximum at 590 nm and a quantum yield of 0.7 [192]. The fluorescence properties of this dye in a range of polymer matrices have been investigated with a view to its use in solar light collection [193]. Some triazole derivatives of stilbene are fluorescent direct dyes suitable for application to cellulosic fibres, an example being compound **69** (CI Direct Yellow 96) [194].

Some cationic methine dyes of the type traditionally used to dye acrylic fibres, such as CI Basic Red 13 (**70a**) and CI Basic Violet 7 (**70b**), are used to an extent in daylight fluorescent pigments. The related dye **71** is described as fluorescent orange, while some aminobenzotriazolyl derivatives, such as **72** [195], and the diphenylpyrazoline derivative **73** [196] are reported as fluorescent yellows. Astrafloxin (**74**, CI Basic Red 12), a cyanine dye, may be used to produce bright pink daylight fluorescent pigments although it has inferior light stability compared with the rhodamines.

Various biscyanovinyl fluorescent yellow dyes, such as compound **75** (R = 1-naphthyl) are reported for use in the bulk dyeing of polymers and synthetic fibres [197].



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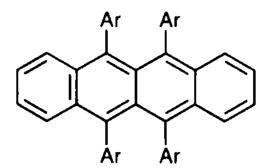

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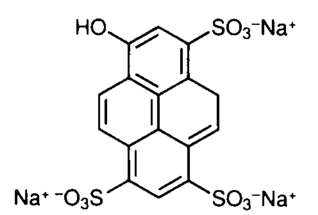

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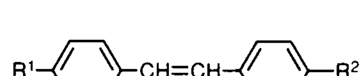
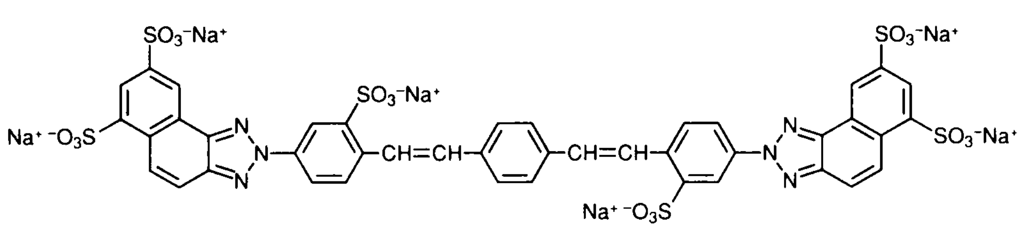


68

	R¹	R²
68a	H	H
68b	NO_2	$N(CH_3)_2$



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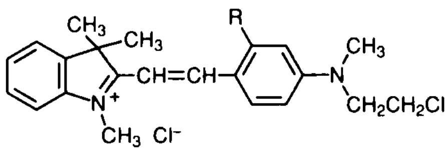
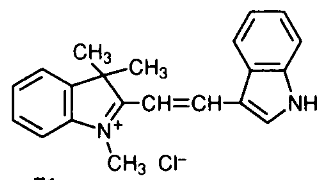


70

70a R = H
70b R = CH_3



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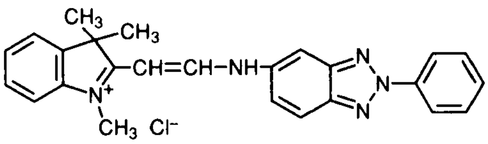

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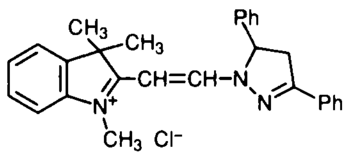

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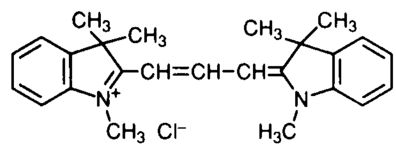

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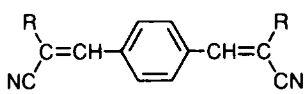

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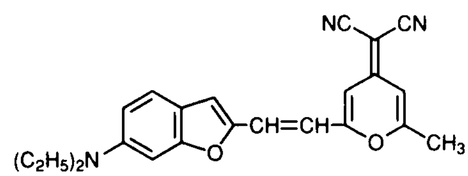


76

Related derivatives with heterocyclic side groups are reported as providing brilliant fluorescent yellow to orange shades when incorporated into polymeric materials [198]. Some orange to violet benzofuran-based methine dyes, of which compound **76** is a representative example, are reported as highly efficient fluorescent materials for application in organic electroluminescent cells [199].

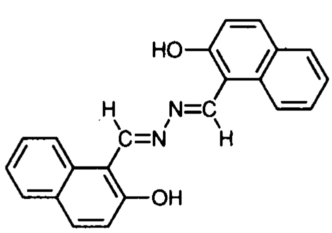
MISCELLANEOUS FLUORESCENT DYES

Several azomethines exhibit yellow-green fluorescence, both in solution and in the solid state. The most important example of a commercial product of this type is the azine **77** (CI Pigment Yellow 101), used as both as a traditional colorant and in flaw detection.

A few polycyclic heterocyclic compounds that have not been included in the previous sections of this review are reported as fluorescent greenish-yellow dyes for polyester. These include quinoxaline derivatives, such as **78**, which have only fair dyeing properties [200]. CI Disperse Yellow 139 (**79**) is a structurally more complex heterocyclic dye that provides bright greenish-yellow shades on polyester, cellulose acetate and polyamides with excellent fastness to light, washing and sublimation [161,201].

The fused benzimidazole derivative **80** is one of a series of fluorescent yellow dyes reported in a recent publication, although they appear to exhibit dyeing properties on polyester that are inadequate for commercial exploitation [202]. An interesting recent development in fluorescent dye chemistry is the report of the synthesis of 1,7-diazaperylene (**81**) and a range of substituted derivatives, which exhibit intense orange-red fluorescence [203,204]. Compound **81** gives an absorption maximum in chloroform of 441 nm, which is a bathochromic shift of only 4 nm with respect to perylene (**36**), its parent hydrocarbon. However, the compound shows an emission maximum at 584 nm, which is some 33 nm bathofluoric with respect to perylene. The large Stokes shift and high quantum yield (0.84) exhibited by compound **81** indicates application potential, for example, in dye lasers and solar collectors. It will be interesting to follow future developments in aza analogues of perylene dyes.

A range of luminescent compounds with two fluorophores have been described [205]. Examples of such bifluorophoric compounds are the fluorescent dyes **27** and **34**, discussed above.



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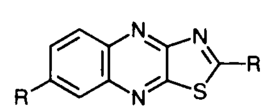

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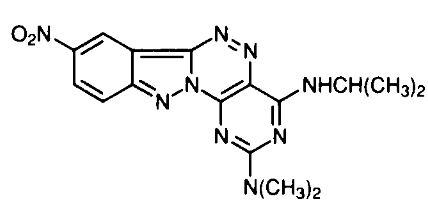

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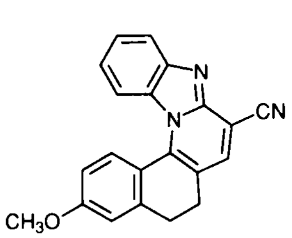

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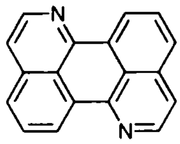

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APPLICATIONS OF FLUORESCENT DYES

In this section a brief account of the most important applications of fluorescent dyes is presented. The reader is directed to the references given for a more detailed account of specific applications, most of which have been previously extensively reviewed.

Daylight fluorescent pigments

Daylight fluorescent pigments consist essentially of fluorescent dyes dissolved in a transparent and colourless polymer. The resulting solid solutions are then ground to a fine particle size for incorporation as pigments into paints, printing inks or plastics by a dispersion process. In these application media the pigments in daylight give rise to colours that possess a remarkable vivid brilliance as a result of the extra glow of fluorescent light. The applications of daylight fluorescent pigments are generally associated with their extremely high visibility and their ability to attract attention. Advertising offers one of the most important applications of fluorescent colours in printing inks, e.g. in posters, magazines and packaging. Many of the important paint applications are in the field of safety. Fluorescent colours are particularly effective on vehicles such as fire engines, ambulances, rescue vehicles and for the marking of military aircraft. On an overcast day a vehicle highlighted with a fluorescent paint is much more visible than the same vehicle painted with a conventional colour.

Safety clothing also very often makes use of daylight fluorescent pigments for reasons of enhanced visibility. The brilliance of the pigments is also responsible for their use for decorative effect in a wide range of plastic products such as toys, bottles and other plastic containers.

Today daylight fluorescent pigments are mostly based on a toluenesulphonamide–melamine–formaldehyde resin matrix, although polyurethane, polyamide and polyester types have also been proposed, particularly when enhanced heat stability is a requirement as in many thermoplastic applications.

Fluorescent dyes most commonly used in the pigments are the red to violet rhodamines **61a** and **61b**, the yellow aminonaphthalimides, notably compound **18**, and benzazolylcoumarins **2a–2d** and **6**. However, a much wider range of other types of fluorescent dyes, including benzothioxanthenes, anthraquinones and methines, may be used, as has been highlighted earlier. A significant problem with many fluorescent pigments is the quenching of the fluorescence as the concentration of the dye in the resin is increased. This is particularly significant with the rhodamines, which exhibit considerable concentration quenching above about 1%. Yellow dyes, on the other hand, may be used to 5 or even 10% in certain matrices before an excessive dulling effect, characteristic of this type of quenching, occurs.

In contrast, mixtures of certain dyes may produce a fluorescence more brilliant than one dye alone, as a result of transfer of energy from one dye to another. In the same way certain fluorescent brightening agents may act as sensitisers enhancing the fluorescence yield from yellow dyes in daylight fluorescent pigments. The light fastness of daylight fluorescent pigments is inferior to that of high-performance organic pigments and this limits to a certain extent their use in exterior applications. Nevertheless, careful selection of the resin matrix and the use of a UV absorbing overlayer can produce a fluorescent colorant with reasonable light stability. It is possible that the development of new dyes and improved resins will make possible far more durable coatings in the future.

The azine **77** is an unusual example of a commercial pigment that exhibits strong fluorescence in its pure solid-state form. Its applications are limited, however, by its relatively poor light stability. It is possible that more durable pigments exhibiting solid state fluorescence, such as the recently patented diketopyrrolopyrrole **55**, will be commercially exploited in the future.

The topic of daylight fluorescent pigments has been discussed in terms of the types of dye used, the composition of the polymer matrix, the manufacturing processes used and the applications of the products in previous reviews and articles [7–10,206].

Dyes for textiles and plastics

The earliest commercial exploitation of fluorescent dyes was the use of rhodamines and related products to dye silk with unusually brilliant colours. A few fluorescent dyes suitable for application to natural protein and cellulosic fibres are available, e.g. CI Direct Yellow 96 (**69**) and CI Reactive Yellow 78 (**24**), but their use is rather limited today.

Of much greater importance are fluorescent dyes suitable for application to synthetic fibres such as polyester, polyamides and polyacrylonitrile. These dyes are used on the appropriate textile fabrics when there is a desire for the particularly striking colours for aesthetic, fashion or functional reasons. They may also be used in non-traditional applications, a familiar example of which is fluorescent yellow tennis balls. Most patents on fluorescent textile dyes have centred on the disperse dye class. These are nonionic molecules of relatively small molecular size applied mainly to polyester fibres, and to a lesser extent to polyamides and cellulose acetates. A wide range of chemical classes of dye suitable for this application have been discussed earlier in this article and previous reviews [7,207].

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Commercial products whose structures have been disclosed include the coumarin CI Disperse Yellow 232 (**2d**), the aminonaphthalimide CI Disperse Yellow 11 (**18**), the benzothioxanthone CI Disperse Red 303 (a mixture of isomers **49** and **50**) and the heterocyclic dye CI Disperse Yellow 139 (**79**).

Also of commercial significance are some water-soluble cationic dyes that are suitable for application to acrylic fibres. Commercial structures that have been disclosed include the coumarin CI Basic Yellow 40 (**6**) and certain methine dyes, including CI Basic Red 13 (**70a**), CI Basic Violet 7 (**70b**) and CI Basic Red 74 (**74**).

Some fluorescent dyes may be used directly in the mass coloration of plastics, notably polystyrene, polymethylmethacrylate and polyesters which exhibit some affinity for disperse or solvent dyes. For these applications there is frequently a requirement for a considerable degree of stability towards heat and, to a certain extent, light. Where there is such a requirement, dyes of the thioindigo and perylene classes are often the materials of choice.

Dyes for lasers

Lasers based on fluorescent dyes have been gaining in importance in recent years. Several inorganic lasers have been developed which emit in the ultraviolet, visible and into the near and far infrared regions of the electromagnetic spectrum. However, these lasers suffer from the drawback that they emit at only a few selected wavelengths in very narrow bands. In contrast dye lasers, as they are generally referred to, emit a broad band of wavelengths, offering the advantage of tunability through a wide range of wavelengths.

The term laser is an acronym for 'light amplification by stimulated emission of radiation'. Having absorbed a quantum of light, a dye molecule is promoted from its ground state S_0 to higher sublevels of the first excited state $S_1{}^*$. Lasing occurs when incident radiation interacts with the dye molecule in its excited state, thereby causing (stimulating) the molecule to decay to the ground state by emission of radiation (fluorescence). In order for the lasing effect to occur, a situation has to be brought about in which the dye molecules exist predominantly in the excited state. This population inversion is achieved by 'pumping' the system with a source of energy, generally a powerful inorganic laser. In contrast to spontaneous emission, the stimulated emission of radiation from dye lasers is strictly coherent (same phase and polarisation) and is of high intensity. These lasers find use in areas such as communications technology, microsurgery, spectroscopy, photochemistry, studies of reaction kinetics, isotope separation and microanalysis.

Many dyes have been investigated for their suitability as laser dyes, the general requirements for which are:

(a) Strong absorption at the excitation wavelength
(b) Minimal absorption at the lasing wavelength, so that there should be as little overlap as possible between absorption and emission spectra
(c) High quantum yield (0.5–1.0)
(d) Short fluorescence lifetime (5–10 ns)
(e) Low absorption in the first excited state at the pumping and lasing wavelengths
(f) Low probability of intersystem crossing to the triplet state
(g) Good photochemical stability.

Fluorescent dyes are used in lasers most commonly in solution in an appropriate solvent. By appropriate dye selection it is possible to produce coherent light of any wavelength from 320 to 1200 nm. For shorter wavelengths (up to approx. 470 nm) aromatic hydrocarbons, such as anthracenes and polyphenyls, and fluorescent brightening agent-type materials, such as stilbenes, oxazoles, coumarins and carbostyrils, are most commonly used. Acridines and benzazolylcoumarins are important in the 470–550 nm range, while rhodamines and related xanthene derivatives are of prime importance in the 510–700 nm region of the visible spectrum. For lasers operating at longer wavelengths into the near-infrared region, oxazines and polymethine (cyanine) dyes are most important.

Several reviews and articles on dye lasers, including the physics and technology of their operation and the properties of the dyes used, have been published previously [7,208–216].

Flaw detection

One of the major areas of application of fluorescent dyes is their use as indicators in the inspection of materials for defects. The timely and reliable detection of defects of components and assemblies in machines operating under heavy-duty conditions, e.g. in aviation and spacecraft, is clearly of critical importance. When dissolved in a penetrating liquid, fluorescent dyes can enhance the contrast against a non-luminescent background of tiny cracks and other defects not necessarily detectable with the naked eye. Metal, glass, ceramics, plastics and other materials can all be examined in this way.

The properties of fluorescing materials required for these applications are:

(a) Intense eye-catching fluorescence in thin films
(b) Good solubility in the solvents used
(c) Non-toxicity
(d) Low cost.

Fluorescent brightening agents may be used for flaw detection purposes but also of considerable importance are xanthene dyes such as fluorescein (**59**), the rhodamines (**61**) and azine **77**.

Analytical applications

The use of fluorescent reagents in analytical methodology is long established. The most important analytical applications of fluorescent dyes are as reagents in quantitative fluorimetric determinations and as fluorescent indicators [7,16]. Fluorimetric determination methods have been established for more than 50 elements

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of the periodic table. The techniques are characterised by the relative simplicity of the instrumentation and a high sensitivity, allowing accurate quantitative determinations of trace amounts of materials to be made. Quantitative fluorimetric analysis is based on the proportionality of the fluorescence intensity of a sample to the amount of the substance being determined.

The measurement of fluorescence intensity may be preceded by interaction of the fluorescent dye with the component of interest. In an important method, ionic associates between fluorescent dye cations, most commonly rhodamines or acridines, and complex anions derived from the substance in question (e.g. $AgBr_2^-$ and $SbCl_6^-$) are formed in aqueous solution. Conversely complex cations may interact with fluorescent dye anions, such as are formed by fluorescein and its derivatives. These associates may be extracted into an appropriate organic solvent, while the excess unreacted fluorescent dye remains in the aqueous phase. The fluorescence intensity of the extract is measured under carefully controlled conditions, and the amount of the element of interest determined from an appropriate calibration curve.

Another method involves the formation of a metal complex of a dye, e.g. from reaction of hydroxyanthraquinone derivatives with a metal ion. As a result of complex formation, a change in the position of the emission spectrum of the dye occurs, manifested as a dramatic increase in its fluorescence intensity, which may be used as a means of quantitative analysis. Other methods utilise the quenching of fluorescence, e.g. as a consequence of dye oxidation.

Fluorescent dyes may be used as indicators in titrations when an end point is signalled by a change in the fluorescence of dye. This technique is especially useful when the titration involves turbid liquids. Fluorescent indicators covering a wide range of pH values are available for use in acid–base titrations. As an example, acridine orange undergoes a change from orange to green fluorescence in the pH range 8.4–10.4. Rhodamines and fluorescein can be used as indicators in redox titrations.

Biological and medical applications

The use of fluorescent dyes in biology and medicine is an immense and growing field. Although well established fluorescent dyes are frequently used, it is sometimes necessary to synthesise coloured derivatives for specific purposes. It is a topic that has been extensively discussed in previous publications [7,215,217–222], and this small section can do no more than present a brief outline of the important applications.

The methods that employ fluorescent dyes are capable of providing high specificity and sensitivity. They generally involve the binding of the dye molecules to substances, tissue, cells or micro-organisms, and subsequent detection by fluorimetric or microscopic techniques. The mechanism of binding may involve physical adsorption or covalent bonding, but in many cases is not fully established. Additionally the dyes used must be non-toxic, exhibit a high quantum yield and have a specific emission spectrum, which is sufficiently distinct from any background fluorescence.

Fluorescence in the blue-violet region is commonly inherent in biological systems, so that tracers fluorescing in the orange, red and green regions are most usefully employed. Among the most widely used fluorochromes in biology and medicine are fluorescein, rhodamines, acridines and their derivatives. Fluorescent dyes are used extensively in the detection and identification of biological substances. Examples of these applications, which most commonly have a diagnostic purpose, include:

(a) Detection of specific antibodies
(b) Identification of nucleic acids, proteins, lipids and polysaccharides in cells
(c) Identification of specific nucleotide sequences in DNA
(d) Early detection of cancer cells.

An extension of these applications of fluorescent dyes in qualitative analysis is their use to determine quantitatively a substance present in a biological substrate at very low concentrations. The dyes used tend to be reactive species containing a functional group capable of reacting specifically and stoichiometrically with the substance in question, such as a chlorotriazinyl group capable of reaction with an amino group, the iodoacetyl group for reaction with a thiol group and the amino group which forms Schiff's bases with aldehydes. Additionally the fluorescence intensity must be directly proportional to the quantity of the fluorochrome which, in turn, must be proportional to the quantity of the substance being analysed. Therefore the fluorochrome must not be subject to concentration quenching or other emission spectral changes associated with the environment of the dye.

Specific fluorescent dyes have been used to investigate the structures of complex biological macromolecules, such as DNA, RNA and enzymes. The fluorescence characteristics of tracers that are covalently bound at specific sites in the biopolymer molecules may be used as tools for structure determination, such as providing information on nucleotide sequences in DNA. Variations in the emission characteristics of the bound fluorescent dye are indicative of conformational changes in the biopolymer molecules, and this leads to their use as probes for investigation of biological activity and the mechanism of biological reactions.

Fluorescent dyes are also used extensively in the study of cell membranes. There is a great structural diversity of fluorescent probes for this purpose, most of which share similar characteristics. In addition to appropriate fluorescence properties, the probes contain hydrophobic groups that are responsible for binding with protein or lipid constituents of the membrane. Applications include:

(a) Measurement of intracellular pH
(b) Measurement of lipid phase viscosity
(c) Investigation of lipoprotein structure
(d) Determination of the charge on a membrane surface

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(e) Determination of potential difference and investigation of ion transport across cell membranes.

In recent years the use of fluorescent probes has made considerable inroads into pharmacology and medicine, and research utilising these materials is no doubt set to continue for years to come. Fluorescent probe molecules, structurally similar to hormones and other biologically active molecules, may be used to provide an insight into the mechanism of the biological activity of these substances. Research using such probes, particularly in the study of biomembrane properties, has already led to the development of new diagnostic procedures for certain diseases and it is likely that they will be at the forefront of future methodology in the search for new potentially active substances.

Solar collectors

There has been considerable effort over many years now to develop the means to convert solar energy into electrical energy. The potential advantages of solar power are obvious. It utilises a non-diminishing energy source and suffers little from the global environmental problems inherent in energy generation by nuclear fission or by the combustion of fossil fuels. One of the deficiencies of conventional silicon cells used for solar energy conversion is their low efficiency, particularly at lower wavelengths in the UV and visible regions. Solar collectors utilise the ability of fluorescent dyes to absorb lower wavelengths of light and to re-emit the energy at higher wavelengths to which the solar cells are more sensitive.

In practice solar collectors contain the dye in a thin sheet of plastic, generally polymethylmethacrylate, whose faces and edges are mirrored to channel the emitted radiation by internal reflection to the edge of the sheet containing the solar cell, as illustrated in Figure 2.

The requirements of dyes for this application are:

(a) High fluorescence quantum yield
(b) Compatibility with the plastic material
(c) High light fastness in the polymer matrix
(d) Minimal overlap of the absorption and emission spectra (generally associated with a large Stokes shift)
(e) High purity.

A wide range of dye classes have been examined for their suitability for application in solar collectors, especially those with appropriate photostability [212,216,223–226]. Coumarins and perylenes are currently the most important classes for these applications.

Other applications

There has been some recent work in the phenomenon of electroluminescence. Organic electroluminescent materials emit light as a result of reversible oxidation/reduction processes in electrode reactions. Devices have been described that comprise an organic electroluminescent component and a fluorescent dye which absorbs the light emitted by the electroluminescent

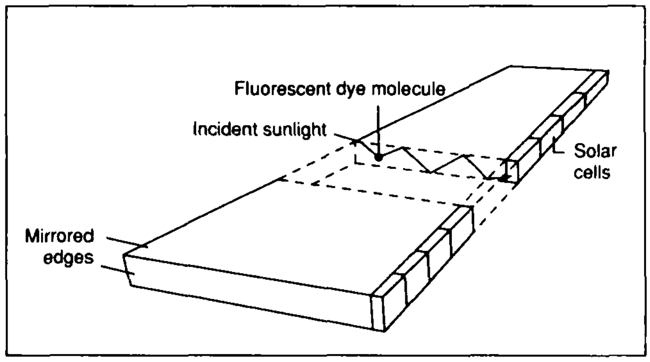

Figure 2 Design of a solar collector using a fluorescent dye

component and re-emits it as visible fluorescent light [227]. Potential applications for such materials include colour display systems for electronic devices.

Fluorescent dyes, particularly of the perylene type, may be used in liquid crystal display systems as a means of enhancing the brightness and visibility of the displays [228]. In geology and hydrology water-soluble fluorescent dyes, notably fluorescein and uranine, are used in tracking water currents [229]. Other applications of fluorescent dyes include fluorescent markers, temperature indicators, radiation dosimetry, map reproduction and security printing.

It is certain that the special characteristics of coloured materials that have the ability to absorb and emit light will continue to excite and fascinate chemists into the future. New applications will emerge that will make use of existing dyes as well as requiring the design and synthesis of new molecules with special characteristics.

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